(iii) The statement "**If $P(x)=0$ is a polynomial equation such that whenever α is a root, $\frac{1}{\alpha}$ is also a root, then the polynomial equation $P(x)=0$ must be a reciprocal equation**" is not true. For instance $2x^3-9x^2+12x-4=0$ is a polynomial equation whose roots are $2, 2, \frac{1}{2}$. Note that $x^3P\left(\frac{1}{x}\right)\neq\pm P(x)$ and hence it is not a reciprocal equation. Reciprocal equations are classified as Type I and Type II according to $a_{n-r}=a_r$ or $a_{n-r}=-a_r$, $r=0, 1, 2,...n$. We state some results without proof :

- For an odd degree reciprocal equation of Type I, $x=-1$ must be a solution.
- For an odd degree reciprocal equation of Type II, $x=1$ must be a solution.
- For an even degree reciprocal equation of Type II, the middle term must be 0. Further $x=1$ and $x=-1$ are solutions.
- For an even degree reciprocal equation, by taking $x+\frac{1}{x}$ or $x-\frac{1}{x}$ as y, we can obtain a polynomial equation of degree one half of the degree of the given equation ; solving this polynomial equation, we can get the roots of the given polynomial equation.

As an illustration, let us consider the polynomial equation

$$6x^6-35x^5+56x^4-56x^2+35x-6=0$$

which is an even degree reciprocal equation of Type II. So 1 and -1 are two solutions of the equation and hence x^2-1 is a factor of the polynomial. Dividing the polynomial by the factor x^2-1, we get $6x^4-35x^3+62x^2-35x+6$ as a factor. Dividing this factor by x^2 and rearranging the terms we get $6\left(x^2+\frac{1}{x^2}\right)-35\left(x+\frac{1}{x}\right)+62$. Setting $u=\left(x+\frac{1}{x}\right)$ it becomes a quadratic polynomial as $6(u^2-2)-35u+62$ which reduces to $6u^2-35u+50$. Solving we obtain $u=\frac{10}{3}, \frac{5}{2}$. Taking $u=\frac{10}{3}$ gives $x=3, \frac{1}{3}$ and taking $u=\frac{5}{2}$ gives $x=2, \frac{1}{2}$. So the required solutions are $+1, -1, 2, \frac{1}{2}, 3, \frac{1}{3}$.

Example 3.27

Solve the equation $7x^3-43x^2=43x-7$.

Solution

The given equation can be written as $7x^3-43x^2-43x+7=0$.

This is an odd degree reciprocal equation of Type I. Thus -1 is a solution and hence $x+1$ is a factor. Dividing the polynomial $7x^3-43x^2-43x+7$ by the factor $x+1$,we get $7x^2-50x+7$ as a quotient. Solving this we get 7 and $\frac{1}{7}$ as roots. Thus $-1, \frac{1}{7}, 7$ are the solutions of the given equation. ■

Example 3.28

Solve the following equation: $x^4-10x^3+26x^2-10x+1=0$.

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